variety

10. RECTIFIERS

A **rectifier** is *a circuit, which uses one or more diodes to convert A.C. voltage into pulsating D.C. voltage.*

A rectifier may be broadly categorized in the following *two types* :

1. Half-wave rectifier and
2. Full-wave rectifier.

10.1. Half-wave Rectifier

Fig. 93 (*a*) shows a half-wave rectifier circuit. It consists of a single diode in series with a load resistor. A *P-N* junction diode can easily be used as a rectifier because it conducts current only when forward biased voltage is acting and does not conduct when reverse bias voltage is acting.

The input to the half-wave rectifier is supplied from the 50 Hz A.C. supply, whose wave form is shown in Fig. 93 (*b*).

Operation :

When and A.C. voltage source is connected across the junction diode as shown in Fig. 93 (*a*) the *positive half cycle of the input acts as a forward bias voltage* and the output across the load resistance varies correspondingly. The *negative half cycle* of the input acts as a *reverse bias* and practically no current flows in the circuit. The output is, therefore, *intermittent*, *pulsating* and *unidirectional*.

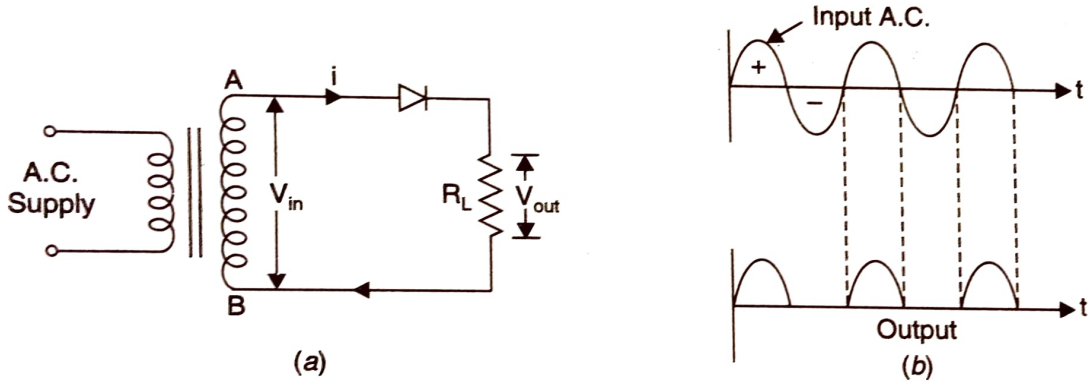


Fig. 93. Half-wave rectifier.

It is evident from the above discussion, that as the *circuit uses only one-half cycle of the A.C. input voltage, therefore, it is popularly known as a "half-wave rectifier".*

Disadvantages :

The main *disadvantages* of a half-wave rectifier are :

(*i*) The A.C. supply delivers power only half the time ; therefore, its *output is low.*

(*ii*) The pulsating current in the load contains alternating component whose basic frequency is equal to the supply frequency. Therefore, an elaborate filtering is required to produce steady direct current.

Efficiency of half-wave rectifier :

The ratio of D.C. power output to the applied A.C. input power is known as ***rectifier efficiency.***

i.e., Rectifier efficiency, $\eta = \dfrac{\text{D.C. power output}}{\text{A.C. input power}}$

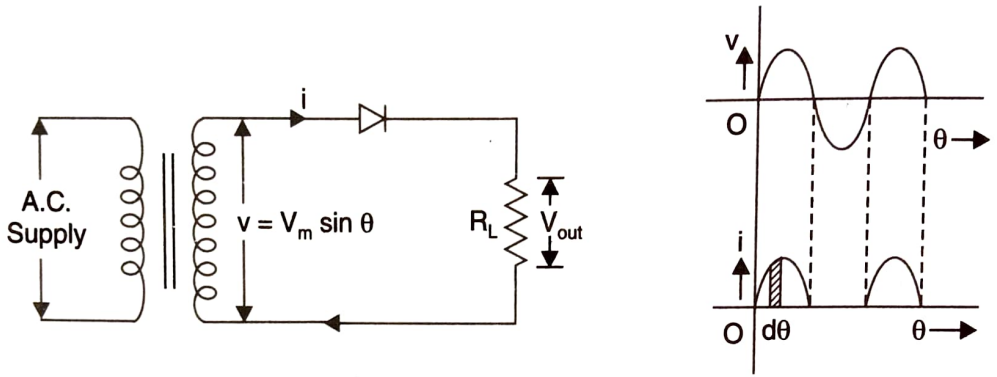


Fig. 94

Consider a half-wave rectifier shown in Fig. 94. Let $v = V_{max} \sin \theta$ be the alternating voltage that appears across the secondary winding.

Let, r_f = Diode forward resistance and

R_L = Load resistance.

D.C. power output :

The output current is pulsating direct current. Therefore, in order to find D.C. power, average current has to be found out,

$$I_{av.} = I_{dc} = \frac{1}{2\pi}\int_0^{\pi} i \, d\theta = \frac{1}{2\pi}\int_0^{\pi} \frac{V_{max} \sin \theta}{(r_f + R_L)} \, d\theta$$

$$= \frac{V_{max}}{2\pi(r_f + R_L)} \int_0^{\pi} \sin\theta \, d\theta = \frac{V_{max}}{2\pi(r_f + R_L)} \Big[-\cos\theta \Big]_0^{\pi}$$

$$= \frac{V_{max}}{2\pi(r_f + R_L)} \times 2 = \frac{V_{max}}{\pi(r_f + R_L)} \times \frac{1}{\pi}$$

$$= \frac{I_{max}}{\pi} \qquad \left[\because \; I_{max} = \frac{V_{max}}{(r_f + R_L)} \right]$$

$\therefore$ D.C. power, $P_{dc} = I_{dc}^2 \times R_L$

$$= \left(\frac{I_{max}}{\pi} \right)^2 \times R_L \qquad ...(i)$$

A.C. power input : The A.C. power input is given by,

$$P_{ac} = I_{rms}^2 \, (r_f + R_L)$$

For a half-wave rectified wave, $I_{r.m.s.} = \frac{I_{max}}{2}$

$\therefore$ $$P_{ac} = \left(\frac{I_{max}}{2} \right)^2 \times (r_f + R_L) \qquad ...(ii)$$

$\therefore$ $$\textit{Rectifier efficiency} = \frac{\text{D.C. output power}}{\text{A.C. input power}}$$

$$= \frac{(I_{max}/\pi)^2 \times R_L}{(I_{max}/2)^2 (r_f + R_L)}$$

i.e., $$\eta_{\text{rectifier(half-wave)}} = \frac{0.406 \, R_L}{r_f + R_L} = \frac{0.406}{1 + \frac{r_f}{R_L}} \qquad ...[25(a)]$$

The efficiency will be maximum when r_f is negligible as compared to R_L.

$\therefore$ *Maximum rectifier (half-wave) efficiency* = **40.6%.**

This means in half-wave rectification, a maximum of 40.6% of A.C. power is converted into D.C. power.

D.C. power.

Example 32. *A diode crystal having internal resistance $r_f = 25\ \Omega$ is used for half-wave rectification. If the applied voltage $v = 60 \sin \omega t$ and load resistance = 725 Ω, find :*

(i) I_{max}, I_{dc}, I_{rms}. *(ii) A.C. input power and D.C. output power.*

(iii) D.C. output voltage. (iv) Efficiency of rectification.

Solution. *Given :* $r_f = 25\ \Omega$; $R_L = 725\ \Omega$; $v = 60 \sin \omega t$.

(i) **I_{max}, I_{dc}, I_{rms} :**

Now, $v = 60 \sin \omega t$ ...(Given)

i.e., $V_{max} = 60$ V

$\therefore$ $$I_{max} = \frac{V_{max}}{r_f + R_L} = \frac{60}{25 + 725} = 0.08 \text{ A or } \mathbf{80\ mA.} \quad \textbf{(Ans.)}$$

$$I_{dc} = \frac{I_{max}}{\pi} = \frac{60}{\pi} = \mathbf{25.46\ mA.} \quad \textbf{(Ans.)}$$

$$I_{rms} = \frac{I_{max}}{2} = \frac{80}{2} = \mathbf{40\ mA.} \quad \textbf{(Ans.)}$$

(*ii*) **A.C. input power, D.C. output power :**

$$P_{ac} = (I_{rms})^2 \times (r_f + R_L) = \left(\frac{40}{1000}\right)^2 \times (25 + 725) = \mathbf{1.2\ W.} \quad \textbf{(Ans.)}$$

$$P_{dc} = (I_{dc})^2 R_L = \left(\frac{25.46}{1000}\right)^2 \times 725 = \mathbf{0.47\ W.} \quad \textbf{(Ans.)}$$

(*iii*) **D.C. output voltage, V_{dc} :**

$$V_{dc} = I_{dc} \times R_L = \frac{25.46}{1000} \times 725 = \mathbf{18.46\ V.} \quad \textbf{(Ans.)}$$

(*iv*) **Efficiency of rectification :**

$$\text{Efficiency of rectification} = \frac{P_{dc}}{P_{ac}} = \frac{0.47}{1.2} = 0.3917 \text{ or } \mathbf{39.17\%.} \quad \textbf{(Ans.)}$$

Example 33. *A half*

10.2. Full-Wave Rectifier

A ***full-wave rectifier*** *is a circuit, which allows a unidirectional current to flow through the load during the entire input cycle.* This can be achieved with *two diodes working alternately.* For the positive half-cycle of input voltage, one diode supplies current to the load and for the negative half-cycle, the other diode does so ; *current being always in the same direction through the load.*

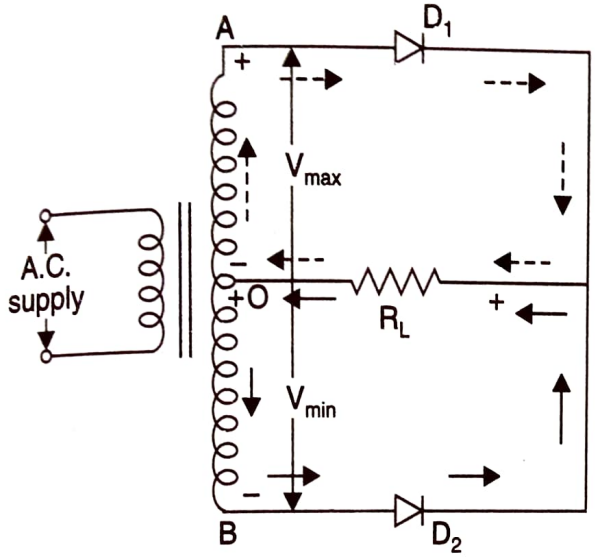
For full-wave rectification the following *two circuits* are commonly used :

1. Centre-tapped full-wave rectifier
2. Full-wave bridge rectifier.

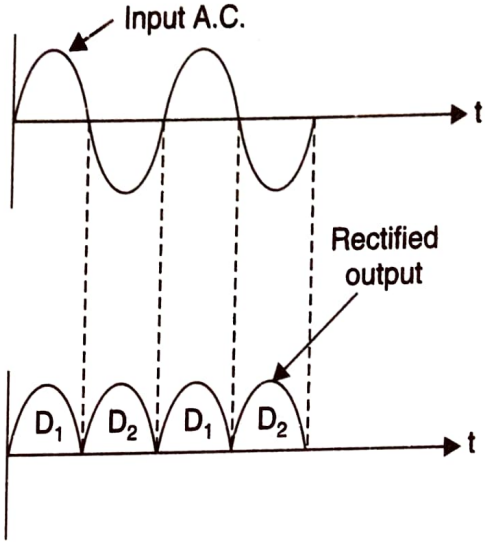
10.2.1. Centre-tapped full-wave rectifier. Fig. 95 shows the circuit of a centre-tapped full-wave rectifier. The circuit uses two diodes (D_1 and D_2) which are connected to the centre-tapped secondary winding AB of the transformer.

Operation :

- During the *positive half-cycle* of secondary voltage, the end A of the secondary winding is positive and end B negative. This makes the diode D_1 forward biased and diode D_2 reverse biased. Therefore, diode D_1 conducts while diode D_2 does not. The conventional current flows through diode D_1, load resistor R_L and the upper half of secondary winding as shown by the dotted arrows.
- During the *negative half-cycle,* the end A of the secondary becomes negative and end B positive. Therefore, D_2 conducts while diode D_1 does not. The conventional current flow is through D_2, R_L and lower half winding as shown by solid arrows.



(*a*)



(*b*) Wave forms of full-wave rectifier.

Fig. 95. Centre-tapped full-wave rectifier.

It may be noted [Fig. 95 (*a*)] that the current in the load R_L is in the *same direction for both the cycles* of input *A.C.* voltage. Therefore, D.C. is obtained from the load R_L.

Also, Peak inverse voltage (*PIV*) = Twice the maximum voltage across the half-secondary winding

i.e., $$PIV = 2\ V_{max}.$$

Advantages :

1. The D.C. output voltage and load current values are twice than those of half-wave rectifiers.
2. The ripple factor is much less (0.482) than that of half-wave rectifier (1.21).
3. The efficiency is twice that of half-wave rectifier.

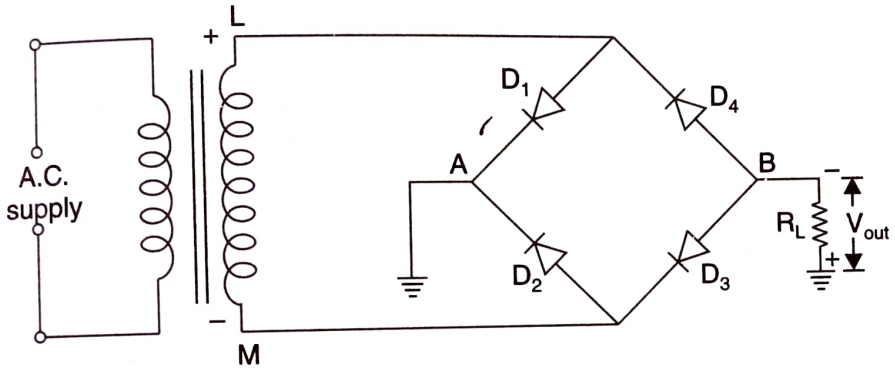
For a full-wave rectifier, the maximum possible value of efficiency is 81.2% while that of half-wave rectifier is 40.6%.

Disadvantages :

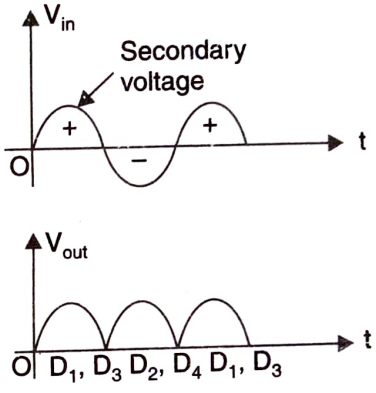
1. The diodes used must have high peak inverse voltage.
2. It is difficult to locate the centre tap on the secondary winding.
3. The D.C. output is small as each diode utilises only one-half of the transformer secondary voltage.

voltage.

10.2.2. Full-wave bridge rectifier. It uses four diodes (D_1, D_2, D_3, D_4) across the main supply, as shown in Fig. 96 (a). The A.C. supply to be rectified is applied to the diagonally opposite ends of the bridge through the transformer. Between other two ends of the bridge, the load resistance R_L is connected.



(a)

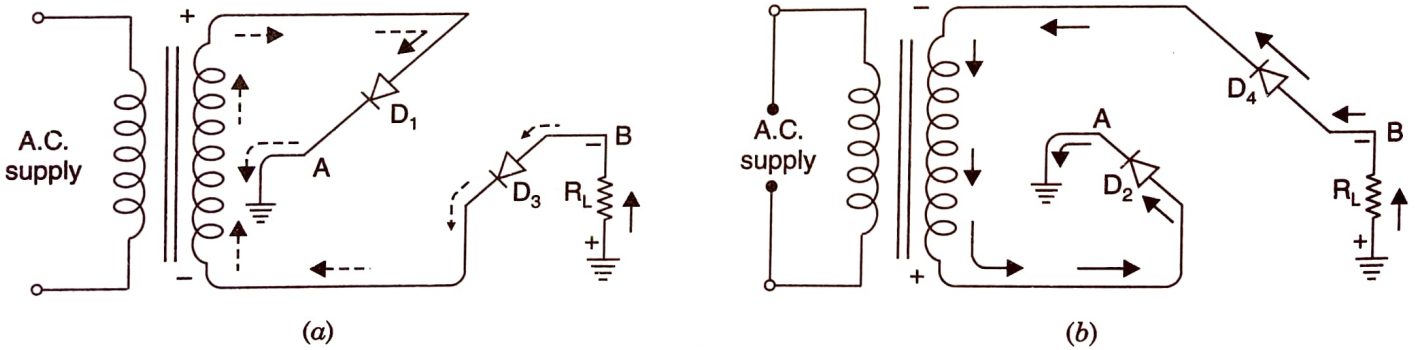


(b)

Fig. 96. Full-wave bridge rectifier.

Operation :

- During *positive half-cycle* of secondary voltage, the end L of the secondary winding becomes positive and end M negative. This makes D_1 and D_3 forward biased while diodes D_2 and D_4 are reverse biased. Therefore, only dioded D_1 and D_3 conduct. These two diodes will be in series through the load R_L as shown in Fig. 97 (a). The current flows (dotted arrows) from A to B through R_L.



(a) (b)

Fig. 97

- During the *negative half-cycle* of the secondary voltage, end L becomes negative and M positive. This makes D_2 and D_4 forward biased whereas diodes D_1 and D_3 are reverse biased. Therefore, only diodes D_2 and D_4 conduct. These two diodes will be in series with R_L as shown in Fig. 97 (b). The current flows (*solid arrows*) from A to B through R_L *i.e.,* in the same direction as for positive half-cycle. Therefore, D.C. output is obtained across R_L.

Further it may noted that *peak inverse voltage (PIV) of each diode is equal to the maximum secondary voltage of transformer.*

Advantages :

1. It can be used with advantage in applications allowing floating input terminals *i.e.,* no output terminal is grounded.

2. The transformer is less costly as it is required to provide only half the voltage of an equivalent centre-tapped transformer used in a full-wave rectifier circuit.

3. No centre-tap is required on the transformer.

4. The output is *twice* that of the centre-tapped circuit for the secondary voltage.

Disadvantages :

1. It uses four diodes as compared to two diodes for centre-tapped full wave rectifier.

2. Since during each half-cycle of A.C. input two diodes that conduct are in series, therefore, voltage drop in the internal resistance of the rectifying unit will be twice as great as in the centre-tapped circuit. This is objectionable when secondary voltage is small.

- These days, the *bridge rectifiers are so common that manufacturers* are packing them as a single unit with bakelite or some other plastic encapsulation with external connections brought out.

Efficiency of full-wave Rectifier :

Fig. 98 shows the *process of full rectification.*

The instantaneous current i is given by

$$i = \frac{v}{(r_f + R_L)} = \frac{V_{max} \sin \theta}{(r_f + R_L)}$$

where r_f and R_L are the diode forward resistance and load resistance respectively.

D.C. output power (P_{dc}) :

$$I_{dc} = \frac{2I_{max}}{\pi}$$

$\therefore$ D.C. power output,

$$P_{dc} = I_{dc}^2 \times R_L = \left(\frac{2I_{max}}{\pi}\right)^2 \times R_L \qquad ...(26)$$

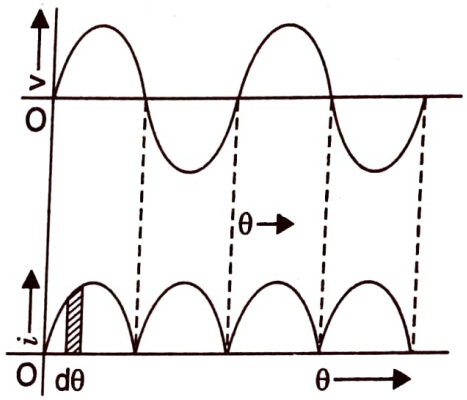


Fig. 98

A.C. input power (P_{ac}) :

$$P_{ac} = (I_{rms})^2 (r_f + R_L)$$

For a full-wave rectified wave, we have

$$I_{rms} = \frac{I_{max}}{\sqrt{2}}$$

$\therefore$ $$P_{ac} = \left(\frac{I_{max}}{\sqrt{2}}\right)^2 (r_f + R_L) \qquad ...(27)$$

$\therefore$ Full-wave rectification efficiency,

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{(2I_{max}/\pi)^2 R_L}{(I_{max}/\sqrt{2})^2 (r_f + R_L)}$$

$$= \frac{8}{\pi^2} \times \frac{R_L}{(r_f + R_L)} = \frac{0.812 R_L}{(r_f + R_L)}$$

i.e.,

$$\eta = \frac{0.812}{1 + \frac{r_f}{R_L}} \qquad ...(28)$$

The efficiency will be maximum if r_f is negligible as compared to R_L.

$\therefore$ $$\eta_{max} = 81.2\%$$

This is double the efficiency than that of half-wave rectifier. Therefore, a full-wave rectifier is *twice as effective as a half-wave rectifier.*

twice as effe

Example 34. *A full-wave rectifier uses two diodes, the internal resistance of each diode may be assumed constant at 25 Ω. The transformer r.m.s. secondary voltage from centre tap to each end of secondary is 60 V and load resistance is 750 Ω. Find :*

(i) The mean load current.

(ii) The r.m.s. value of load current.

Solution. *Given :* $r_f = 25\ \Omega$; $R_L = 750\ \Omega$

(*i*) **The mean load current, I_{dc} :**

$$V_{max} = 60 \times \sqrt{2} = 84.85\ \text{V}$$

Maximum load current, $I_{max} = \dfrac{V_{max}}{(r_f + R_L)}$

$$= \frac{84.85}{25 + 750} = 0.109\ \text{A or } 109\ \text{mA}$$

$\therefore$ $$I_{dc} = \frac{2I_{max}}{\pi} = \frac{2 \times 109}{\pi} = \mathbf{69.39\ mA.}\quad \textbf{(Ans.)}$$

(*ii*) **The r.m.s. value of load, I_{rms} :**

$$I_{rms} = \frac{I_{max}}{\sqrt{2}} = \frac{109}{\sqrt{2}} = \mathbf{77.1\ mA.}\quad \textbf{(Ans.)}$$

Example 35. *In the centre-tapped circuit shown in Fig. 99, the diodes are assumed to be ideal (i.e., having zero internal resistance). Find :*

(*i*) *D.C. output voltage.*

(*ii*) *Peak inverse voltage.*

(*iii*) *Rectification efficiency.*

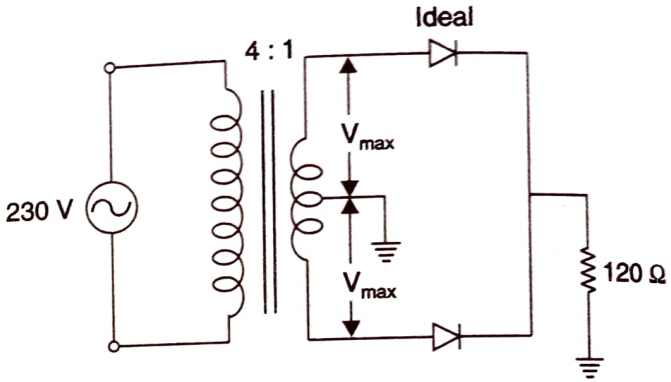


Fig. 99

Solution. *Given :* $\frac{N_2}{N_1} = \frac{1}{4}$; $R_L = 120\ \Omega$;

$r_f = 0\ \Omega$; $V_1 = 230$ V.

(*i*) **D.C. output voltage, V_{dc} :**

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{1}{4}$$

$$\therefore \quad V_2 = 230 \times \frac{1}{4} = 57.5\ \text{V}$$

Maximum voltage across the secondary = $\sqrt{2} \times 57.5 = 81.32$ V

Maximum voltage across half secondary winding,

$$V_{max} = \frac{81.32}{2} = 40.66\ \text{V}$$

Now,
$$I_{dc} = \frac{2V_{max}}{\pi R_L} = \frac{2 \times 40.66}{\pi \times 120} = 0.216\ \text{A}$$

$$\therefore \quad V_{dc} = I_{dc} \times R_L = 0.216 \times 120 = \mathbf{25.92\ V.}\quad \textbf{(Ans.)}$$

(*ii*) **Peak inverse voltage (PIV) :**

$$PIV = \mathbf{81.32\ V.}\quad \textbf{(Ans.)}$$

(*iii*) **Rectification efficiency :**

$$\text{Rectification efficiency} = \frac{0.812}{1 + \frac{r_f}{R_L}} \qquad \text{...[Eqn. (28)]}$$

Since r_f is zero, therefore,

Rectification efficiency = 0.812 or **81.2%.** **(Ans.)**

Example 36. *In the bridge type shown in Fig.*

Rectification efficiency = 0.812 or 81.2%.

Example 36. *In the bridge type shown in Fig. 100, the diodes are assumed to be ideal. Find :*

(i) D.C. output voltage. *(ii) Peak inverse voltage.*

(iii) Output frequency.

Assume primary to secondary turns to be 5.

Solution. *Given :* $\frac{N_2}{N_1} = \frac{1}{5}$; $V_1 = 230$ V ; $r_f = 0$; $R_L = 250\ \Omega$

(i) D.C. output voltage, V_{dc} :

R.M.S. secondary voltage, $V_2 = V_1 \times \frac{N_2}{N_1} = 230 \times \frac{1}{5} = 46$ V. $\left[\because \frac{V_2}{V_1} = \frac{N_2}{N_1}\right]$

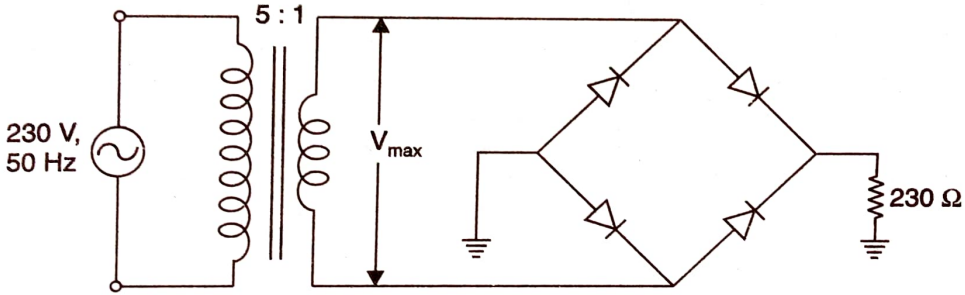


Fig. 100

Maximum voltage across secondary = $46 \times \sqrt{2} = 65$ V

Average current, $I_{dc} = \dfrac{2V_{max}}{\pi R_L} = \dfrac{2 \times 65}{\pi \times 250} = 0.165$ A

$\therefore \quad V_{dc} = I_{dc} \times R_L = 0.165 \times 250 =$ **41.25 V. (Ans.)**

(*ii*) **Peak inverse voltage** *(PIV) :*

The peak inverse voltage is equal to the maximum secondary voltage *i.e.,*

$PIV =$ **65 V. (Ans.)**

(*iii*) **Output frequency :**

In full-wave rectification, there are two output pulses for each complete cycle of the input A.C. voltage. Therefore, the output frequency is *twice* that of the A.C. supply frequency, *i.e.,*

$f_{on} = 2f_{in} = 2 \times 50 =$ **100 Hz. (Ans.)**